

Binary Classification of Traffic Collisions Events in Ottawa

Sheena Verana
School of Advanced Technology
Algonquin College
Ottawa, Canada
vera0016@algonquinlive.com

Andrianna Wardill
School of Advanced Technology
Algonquin College
Ottawa, Canada
ward0366@algonquinlive.com

Olamilekan Elegbede
School of Advanced Technology
Algonquin College
Ottawa, Canada
eleg0003@algonquinlive.com

Abstract— Traffic collisions are a safety concern in the City of Ottawa. Using traffic collision data from the City of Ottawa, this project performs exploratory data analysis (EDA) on the dataset that has 94,406 rows and 28 columns. The original multi-class accident classification was transformed to a binary classification of collisions as fatal or non-fatal. Exploratory Data Analysis was conducted to examine class imbalance, missing data patterns, feature relationships, and temporal and geospatial trends. Data cleaning, feature engineering, and five supervised machine learning models were applied, including Logistic Regression, Random Forest, XGBoost, Gradient Boosting, and Linear SVM. Three resampling techniques (SMOTE, SMOTE-ENN, and ADASYN) were implemented to address severe class imbalance (0.18% fatal). Models were assessed using the Matthews Correlation Coefficient (MCC), F1-score, Precision-Recall AUC, and True Negative Rate. Gradient Boosting with SMOTE on an 85/15 train-test split achieved the highest MCC of approximately 0.10. Subsequent hyperparameter tuning via GridSearchCV did not result in meaningful improvement, showing the fundamental challenge of detecting rare fatal events from limited positive samples. The results demonstrate the potential of machine learning to support traffic safety analysis, as well as its limitations when applied to highly imbalanced datasets.

Keywords—Binary Classification, Traffic Collisions, Data Analysis, Ottawa

I. INTRODUCTION

Ottawa has consistently experienced a high volume of traffic collisions, with approximately 20,000 incidents reported annually according to the Ottawa Police Service. Although collision rates declined in 2020 due to reduced mobility during the COVID-19 pandemic, recent data indicates that they have returned to and even exceeded pre-pandemic levels. For instance, over 3,100 collisions were reported within the first six weeks of 2026 alone [1], including cases involving fatalities, injuries, pedestrian involvement, and hit-and-run incidents. These trends highlight the growing concern around road safety and the increasing demand for emergency response services across the city.

The goal of this project is to develop a supervised machine learning model capable of classifying traffic collisions as fatal or non-fatal based on historical collision data. By leveraging variables such as collision location, time, environmental conditions, type of impact, and involvement of pedestrians or multiple vehicles, the model aims to identify patterns that are strongly associated with fatal outcomes. This binary classification approach simplifies the prediction task while maintaining practical relevance for real-world applications.

This research is important because it has direct implications for emergency response optimization and public safety planning. Predicting the likelihood of a fatal outcome in real time can support more informed decision-making regarding resource allocation and response prioritization. Additionally, the findings from this model can help city planners and policymakers identify high-risk areas and conditions, enabling targeted interventions to reduce fatalities and improve overall road safety in Ottawa.

II. DATA SOURCE

The project used a publicly available dataset from the City of Ottawa Open Data Portal, called Traffic Collision Data [2]. This dataset includes identifiers, spatial coordinates, temporal variables, accident characteristics, road and environmental conditions, traffic control information, counts of involved road users, and detailed injury severity metrics from 2017 to 2024, with a total of 94,406 rows and 28 columns. Table I provides the feature list and corresponding data types.

III. EXPLORATORY DATA ANALYSIS (EDA)

This section presents the exploratory data analysis (EDA) performed on the dataset. The EDA is structured to expose class imbalance, identify missing data patterns, examine feature-severity relationships, quantify correlations, and extract temporal and geospatial insights relevant to the binary classification of fatal and non-fatal collisions.

A. Initial Inspection of Dataset

The dataset contains 94,406 rows and 28 columns after loading it into the Python notebook. The columns can be grouped into four main categories. First, there is geographic information provided in multiple formats, including Web Mercator, MTM Zone 9 /NAD83, and standard latitude/longitude (WGS84), along with a location feature that states the intersection at which the collision happened. Second, there are temporal features such as the accident year and date. Third, the dataset includes several categorical variables describing environmental, road surface, lighting, impact type, and traffic control conditions. Finally, there are numeric columns that capture injury counts, including total, minimal, minor, major, and fatal injuries, as well as the number of vehicles, pedestrians, bicycles and motorcycles involved.

The Accident_Date column is initially stored as an object type and requires conversion to datetime format for temporal feature extraction. Several numerical columns can be converted to integers. The remaining object-typed variables

such as environmental conditions, road surface, lighting, and traffic control, represent categorical features and will need to be encoded before model training.

B. Target Variable and Class Distribution Analysis

The dataset's target variable, `Classification_Of_Accident`, originally contained four different categories. Table I summarizes the original class distribution. For this analysis, these were simplified into two groups for binary classification: Fatal (01 – Fatal injury) and Non-Fatal (02 – Non-fatal injury, 03 – P.D. only, 04 – Non-reportable).

TABLE I. ORIGINAL ACCIDENT CLASSIFICATION DISTRIBUTION

Classification	Count	Percentage (%)
03 – Property Damage Only	76,671	81.21
02 – Non-Fatal Injury	15,002	15.89
04 – Non-Reportable	2,563	2.71
01 – Fatal Injury	170	0.18

The resulting binary distribution is extremely skewed: 170 fatal collisions (0.18%) versus 94,236 non-fatal collisions (99.82%). This indicates a clear class imbalance within the dataset.

When building the model, stratified train/test split should be used to preserve the proportion of the minority class in both sets. Also, oversampling methods such as SMOTE (Synthetic Minority Over-sampling Technique) may be applied to the training data only, in order to avoid data leakage.

Evaluation metrics such as the Precision-Recall Area Under Curve (PR-AUC), Matthews Correlation Coefficient (MCC), true negative rate (TNR), F1-score, precision, recall for the positive (Fatal) class must be selected due to an imbalanced dataset.

The precision is used to measure the proportion of predicted positive cases that are actually positive. This is important to use in the imbalanced dataset to avoid excessive false positives. The recall (sensitivity) is the proportion of actual positive cases correctly identified, it provides focus on detecting the ‘Fatal’ accidents. F1-score, is the harmonic mean of precision and recall. It is a useful metric for identifying the false positives and negatives. The Matthews Correlation Coefficient (MCC) was chosen as it considers all four values of the confusion matrix. It is an important metric when comparing models as it takes the imbalance of the dataset into account when calculating. Finally, the Precision-Recall Area Under Curve (PR-AUC) was chosen as it was designed to focus on the positive class (Fatal) which is important with an imbalanced dataset. Whereas other options like Area Under the Receiver Operating Characteristic Curve (ROC-AUC) can give overly optimistic estimates due to the dominant class (Non-Fatal), which is not beneficial for an imbalanced dataset [3]–[7].

C. Missing Value Analysis

A total of 17 columns out of the 28 columns contain missing values. Participant count columns (such as pedestrians, bicycles, and motorcycles) are only filled in when those participants are involved in a collision. When

values are missing, it typically means none were involved, so replacing them with zero is appropriate. A similar pattern appears in the injury-related columns, where values are only recorded if injuries occurred, while property-damage-only cases naturally have no entries.

The `num_of_fatal` column, which is populated in only 169 records (one fewer than the 170 fatal collision labels due to a single record inconsistency), directly indicates whether a collision was fatal. It effectively reveals the target outcome. Therefore, it must be removed before training the model to avoid data leakage.

D. Feature vs. Severity Analysis

Each categorical and numerical feature was analyzed against the binary `fatal_label` target using cross-tabulation. Both total collision counts and fatality percentages were examined to distinguish between common conditions and those associated with higher risk.

1) Environmental Conditions

Clear weather conditions account for the majority of collisions and fatalities, with 151 of the 170 fatal cases occurring under clear conditions (Fig. 1(a)). However, when considering fatality rates, a different pattern is seen (Fig. 1(b)). Conditions such as fog, mist, smoke, or dust exhibit a fatality rate of 0.71%, and strong winds 0.68%, compared to 0.20% under clear conditions. Drifting snow also exceeds the clear-weather baseline at 0.23%. While most collisions occur under normal conditions, hazardous conditions tend to be associated with higher severity.

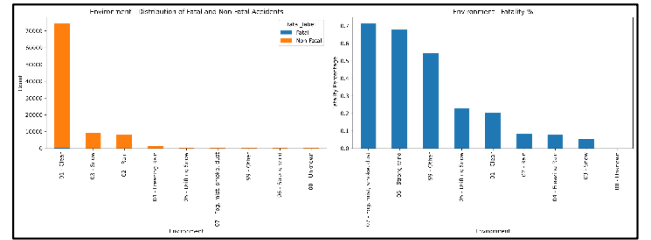


Fig. 1. Environmental conditions: (a) Distribution of fatal and non-fatal collisions, (b) Fatality rate by environmental condition.

2) Road Surface Conditions

Dry road surfaces account for the largest share of both total collisions and fatal cases (133 fatalities), reflecting their prevalence in the dataset (Fig. 2(a)). However, loose sand or gravel surfaces exhibit the highest fatality rate at 1.25%, despite appearing in only 80 records (Fig. 2(b)). Unknown surface conditions also show a relatively high rate of 0.99%. Wet (0.15%) and icy (0.09%) surfaces have lower fatality rates than expected. This may indicate that drivers adjust their behavior in hazardous conditions by reducing speed and exercising greater caution.

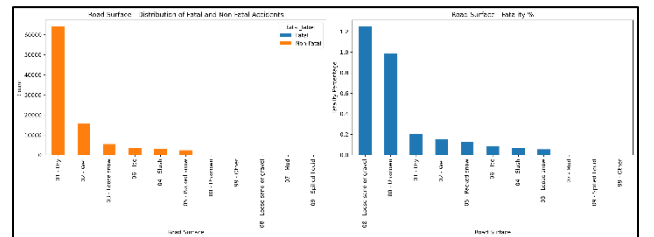


Fig. 2. Road surface conditions: (a) Distribution of fatal and non-fatal collisions, (b) Fatality rate by road surface condition.

3) Lighting Conditions

Most collisions occur during daylight (89 fatal cases), but fatality rates are higher under reduced visibility conditions (Fig. 3(a)). Nighttime conditions show a fatality rate of 0.31%, and dawn 0.30%, compared to 0.14% during daylight (Fig. 3(b)). Dusk conditions also exhibit a slightly higher rate of 0.19%. Reduced visibility contributes to increased collision severity.

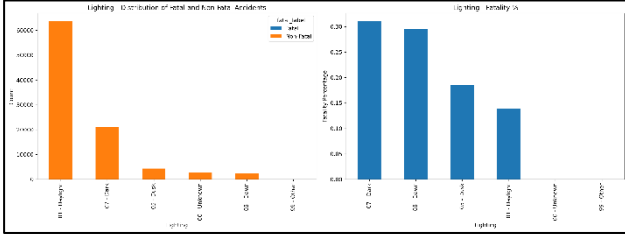


Fig. 3. Lighting conditions: (a) Distribution of fatal and non-fatal collisions, (b) Fatality rate by lighting surface condition.

4) Initial Impact Type

Rear-end collisions are the most common impact type in the dataset (31,720 records in Fig. 4(a)) but have the lowest fatality rate at 0.02% (Fig. 4(b)). Approaching (head-on) collisions show the highest fatality rate at 1.49%, despite being relatively infrequent.

Single motor vehicle (SMV) collisions, which include run-off-road events and impacts with fixed objects, are both common (14,197 records) and relatively severe, with a fatality rate of 0.65%. Turning movement and angle collisions show moderate fatality rates.

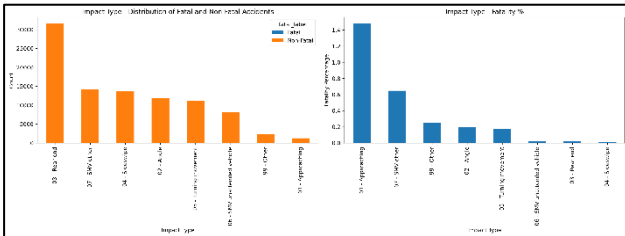


Fig. 4. Impact Type: (a) Distribution of fatal and non-fatal collisions, (b) Fatality rate by impact type.

5) Traffic Control

Locations without traffic control account for the highest number of collisions (43,700 in Fig. 5(a)) and fatalities (102), with a fatality rate of 0.23% (Fig. 5(b)). Stop sign locations are associated with a slightly higher rate of 0.30%, while traffic signals are associated with a lower rate of 0.09%. Traffic gates exhibit the highest fatality rate at 8.33%. However, this is based on only 12 observations and should be interpreted with caution. The results suggest that less regulated or higher-speed environments are associated with increased severity.

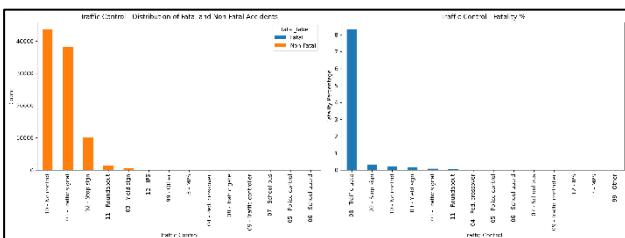


Fig. 5. Traffic Control: (a) Distribution of fatal and non-fatal collisions, (b) Fatality rate by traffic control.

6) Vehicle, Pedestrian, Bicycle, and Motorcycle Involvement

Single-vehicle collisions have a higher fatality rate (0.45%) than two-vehicle collisions (0.10%) as seen in Fig. 6(b), which is consistent with loss-of-control scenarios at higher speeds.

Pedestrian involvement significantly increases fatality risk. Collisions involving one pedestrian have a fatality rate of 2.61%, rising to 3.17% when two pedestrians are involved (Fig. 7(b)). Motorcycle collisions also show elevated risk, with a fatality rate of 2.69% for single motorcycles and 10.00% for two motorcycles (Fig. 8(b)), although the latter is based on a small sample. Bicycle collisions exhibit a moderate fatality rate of 0.61% (Fig. 9(b)). There is vulnerability with these road users.

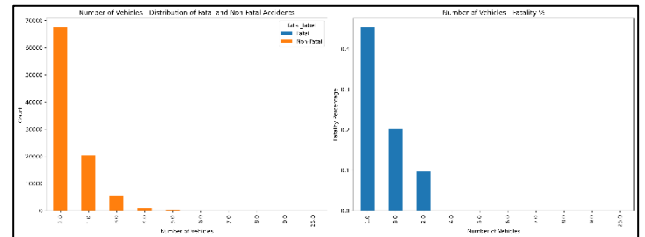


Fig. 6. Number of Vehicles: (a) Distribution of fatal and non-fatal collisions, (b) Fatality rate by number of vehicles.

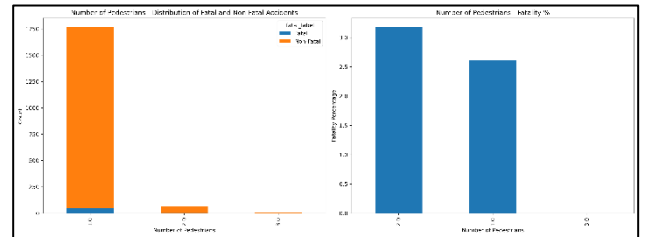


Fig. 7. Number of Pedestrians: (a) Distribution of fatal and non-fatal collisions, (b) Fatality rate by number of pedestrians.

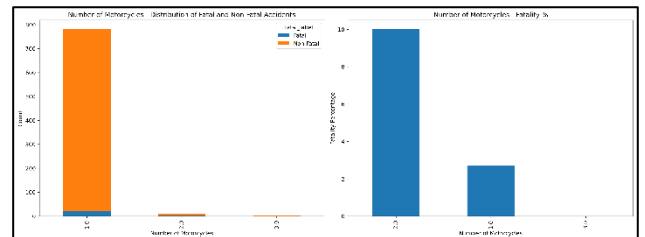


Fig. 8. Number of Motorcycles: (a) Distribution of fatal and non-fatal collisions, (b) Fatality rate by number of motorcycles.

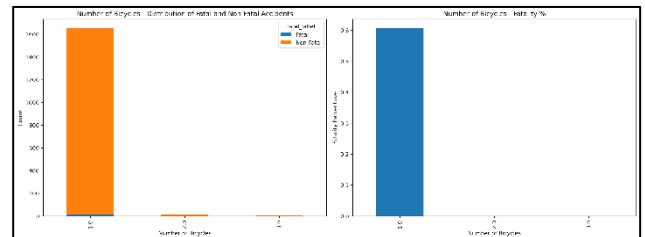


Fig. 9. Number of Bicycles: (a) Distribution of fatal and non-fatal collisions, (b) Fatality rate by number of bicycles.

E. Injury Severity Analysis

Boxplot analysis of injury count distributions, separated by fatal and non-fatal outcomes, reveals clear differences in

how various injury types relate to collision severity as seen in Table II.

Among these features, `num_of_major` stands out as the most informative. Major injuries are cases that typically require hospital admission, so they are closely linked to the same conditions that lead to fatalities. As a result, this feature serves as a strong predictor for severe collisions and is a valuable input for machine learning models.

However, this variable contains a large number of missing values (93,589 records), which must be addressed before modeling. Replacing empty values to zero is appropriate for cases involving property-damage-only collisions, where no injuries are recorded.

The `num_of_minimal` provides the least useful information, as its distribution shows little separation between fatal and non-fatal cases. This feature could be removed during feature selection or dimensionality reduction without significantly affecting model performance.

TABLE II. PREDICTIVE SIGNALS OF FEATURES

Feature	Predictive Signal	Interpretation
<code>num_of_injuries</code>	Moderate	Fatal collisions tend to involve a higher total number of injuries, reflecting greater overall severity.
<code>num_of_minimal</code>	Weak	Minimal injuries show little difference between fatal and non-fatal cases and do not meaningfully distinguish severity.
<code>num_of_minor</code>	Weak–Moderate	Slightly higher in fatal collisions, but there is substantial overlap between the two classes.
<code>num_of_major</code>	Strong	Significantly higher in fatal collisions, making it a strong indicator of severity.
<code>num_of_fatal</code>	Leakage	Directly defines the target and must be excluded from modeling to avoid data leakage.

F. Correlation Analysis

Correlations were computed across all numerical features to identify relationships and potential redundancy. A total of 18 feature pairs showed strong correlations ($|r| \geq 0.70$).

The dataset includes collision location in multiple coordinate systems: Web Mercator (X, Y), MTM Zone 9 (X_Coordinate, Y_Coordinate), and WGS84 (Latitude, Longitude). These representations are highly correlated, with near-perfect correlations ($r \approx 0.99$ – 1.00) between matching coordinate pairs.

All of these columns describe the same location in different formats, keeping all of them introduces unnecessary duplication without adding new information. It is sufficient to retain a single coordinate system.

The `ObjectId` column shows a very strong correlation with `Accident_Year` ($r = 0.98$), which suggests that it is simply an auto-incremented identifier assigned over time. As a result, it does not contain any meaningful information about the collision itself.

Several injury-related features are also strongly correlated with one another. For example, `num_of_fatal` is highly

correlated with `num_of_motorcycles` ($r = 1.00$), likely due to the small number of motorcycle-related fatal collisions in the dataset. It also shows strong correlations with `num_of_minimal` ($r = 0.89$), `num_of_major` ($r = 0.84$), and `num_of_minor` ($r = 0.80$).

These relationships do not necessarily indicate true redundancy, but rather reflect the fact that different types of injuries often occur together in more severe collisions.

G. Temporal Analysis

1) Annual Trend

Collision numbers dropped noticeably in 2020 and 2021, which aligns with the COVID-19 pandemic and the resulting decrease in travel due to public health restrictions. Even though the total number of collisions decreased, the fatality rate reached its highest point in 2021 at approximately 0.24% (Fig. 10).

This suggests that the nature of collisions may have changed during this period. With fewer vehicles on the road, drivers may have traveled at higher speeds, particularly on highways or less congested routes, leading to more severe outcomes. After 2021, collision numbers gradually returned to pre-pandemic levels, while fatality rates stabilized.

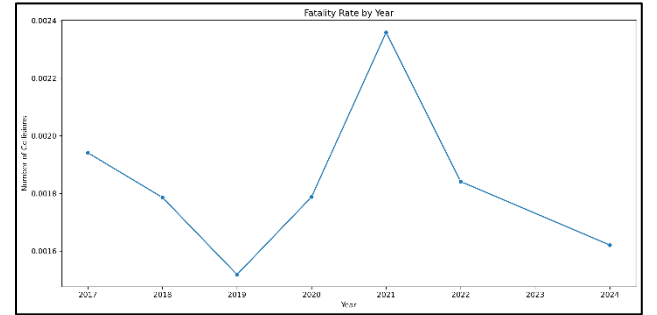


Fig. 10. Fatality Rate by Year.

2) Seasonal and Monthly Patterns

Collision frequency shows a seasonal trend, with the highest number of incidents occurring during the winter months (November to January). This is consistent with Ottawa's climate, where snow, ice, and reduced daylight create more hazardous driving conditions. April has the lowest collision numbers, likely reflecting improved road conditions after winter and relatively lower traffic compared to peak summer months.

However, the pattern for fatality rates differs. The highest fatality rate occurs in July (approximately 0.35% in Fig. 11), with summer months generally showing higher severity despite fewer total collisions. This may be explained by higher driving speeds in clear conditions, increased travel, and greater exposure of road users such as pedestrians and cyclists.

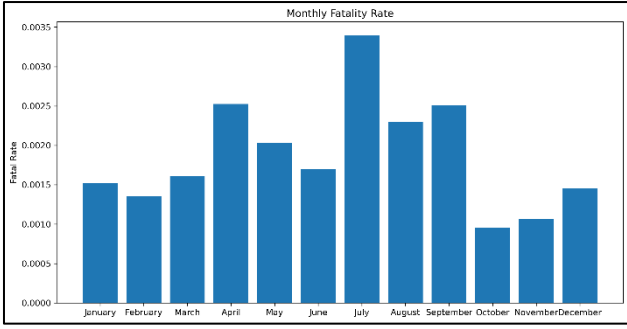


Fig. 11. Fatality Rate by Month.

3) Weekday Patterns

Collision frequency is highest on weekdays, particularly on Thursdays and Fridays, which aligns with typical traffic patterns. The lowest number of collisions occurs on Sundays.

However, fatality rates are highest on weekends, with Sunday showing the highest rate (approximately 0.26% in Fig. 12), followed by Saturday. This suggests that while fewer collisions occur on weekends, they tend to be more severe. Possible explanations include higher travel speeds, lower traffic congestion, and increased likelihood of impaired driving.

The temporal patterns show strong predictive signals. Features such as month, day of week, and season should be included in the machine learning model.

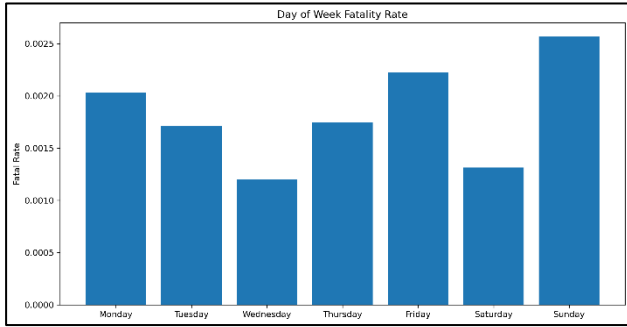


Fig. 12. Fatality Rate by Weekday.

H. Geospatial and Location Hotspot Analysis

A location-based analysis was performed using the Location field to identify collision hotspots across the city. Excluding records with unknown locations (2,076 cases), the highest number of collisions occurs along a segment of Highway 417 between IC124 RAMP65 and IC122 RAMP51 (521 collisions in Fig. 13). Other high-frequency locations include Hunt Club Road at Riverside Drive (334 collisions) and St. Joseph Boulevard at Jeanne d'Arc Boulevard (329 collisions). These results are consistent with expectations, as these areas represent major roads and busy intersections within Ottawa's map.

When focusing specifically on fatal collisions, no single location stands out as a dominant hotspot (Fig. 14). The highest number of fatal collisions at any individual location is only two, observed across several different sites, including segments of Bank Street, Albion Road, Regional Road 174, and Highway 417.

This more dispersed pattern suggests that fatal collisions are not concentrated in specific locations, but are instead

influenced more by situational factors such as collision type, vehicle speed, and the involvement of vulnerable road users.

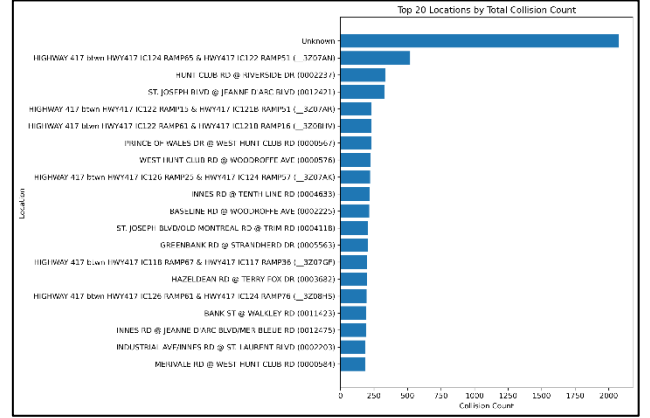


Fig. 13. Top 20 Locations by Total Collision Count.

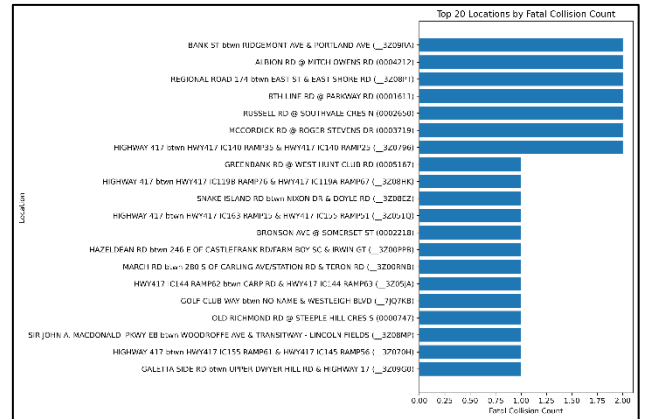


Fig. 14. Top 20 Locations by Fatal Collision Count.

IV. DATA CLEANING AND PREPROCESSING

A. Dataset Description

As mentioned earlier in Section II - Data Sources, the dataset comprises 94,406 traffic collision records spanning 28 attributes. Features include collision contextual variables such as road surface condition, environmental condition, lighting, traffic control type, impact type, and counts of involved road users, along with geospatial coordinates and a four-class accident severity target variable.

B. Duplicate Detection

A preliminary check confirmed the absence of duplicate records across all 94,406 entries; therefore, no records were removed at this stage.

C. Handling Missing Values

An inspection of null values revealed two distinct missing data patterns requiring different treatment strategies.

The first pattern concerned post-incident derived fields: Max_injury, num_of_injuries, num_of_minimal, num_of_minor, num_of_major, and num_of_fatal, which are populated only after injury assessment and accident classification has occurred. Keeping these features would cause data leakage, as they are causally derived from the target variable. Accordingly, all six columns were dropped from the dataset.

The second pattern affected participant count fields: num_of_pedestrians, num_of_bicycles, num_of_motorcycles, and num_of_vehicles. Missing values in these columns reflect the absence of those road user types from the collision rather than unknown information. These fields were therefore imputed with zero.

D. Feature Reduction

Several columns were identified as irrelevant to predictive modeling and removed. Projected coordinate fields (X, Y, X_Coordinate, Y_Coordinate) are redundant given the presence of geographic latitude and longitude. Administrative identifiers (ID, ObjectID, Geo_ID, Location) carry no predictive signals. Accident_Year was removed as temporal granularity is preserved in the full date field. Geospatial fields (Lat, Long) were retained in a supplementary reference table mapping each Geo_ID to its corresponding coordinates for potential future spatial analysis, but excluded from the modeling dataset. Following this step, the dataset was reduced to 11 features.

E. Data Type Standardization

Participant count fields (num_of_vehicles, num_of_pedestrians, num_of_bicycles, num_of_motorcycles) were cast from float64 to integer, reflecting their discrete count nature. The Accident_Date field was parsed from a string representation to a datetime type to enable temporal feature extraction in subsequent stages.

F. Categorical Value Normalization

Categorical features: Initial_Impact_Type, Road_1_Surface_Condition, Environment_Condition_1, Light, and Traffic_Control were stored with numeric prefixes in the source data (e.g., "02 - Angle"). These prefixes were removed, retaining only the descriptive label (e.g., "Angle"), to prevent encoding methods from treating the ordinal prefix as part of the category identity.

G. Binary Target Creation

The original target variable, as shown in Table [x], shows the distribution of the four accident classes in our dataset. To formulate a binary classification task, two decisions were made. First, the Non-reportable class was excluded, as it does not correspond to a definitive injury outcome and introduces ambiguity into the learning objective. Second, Property Damage Only records were merged with Non-fatal injury records to form a unified negative class, as both outcomes share the common characteristic of no loss of life. The resulting binary dataset contains 170 fatal and 91,673 non-fatal records. This imbalance is addressed through targeted resampling and cost-sensitive learning strategies described in Section VI.

V. FEATURE ENGINEERING

Following data preprocessing, a structured feature engineering was applied to the cleaned dataset of 91,843 records across 11 features. The objective was to construct a richer, numerically encoded representation that captures domain-relevant accident risk factors while remaining suitable for binary classification. The final engineered dataset contains 60 features.

A. Target Variable Encoding

The binary target variable was encoded numerically by mapping Fatal injury to 1 and Non-fatal injury to 0, making it compatible with standard supervised learning classification algorithms.

B. Temporal Feature Extraction

The Accident_Date field was broken into four new features to capture time-based patterns in collision severity: calendar month (1–12), day of the week (0–6, where 0 represents Monday), a binary weekend indicator, and a categorical season variable encoding winter (0), spring (1), summer (2), and fall (3). The original date column was subsequently dropped, as its information is fully represented by the extracted features.

C. Domain Knowledge Features

Six binary and aggregate features were constructed based on established traffic safety principles:

1) Vulnerable Road User Exposure

Two features were derived to quantify the involvement of road users at elevated fatality risk: pedestrians, cyclists, and motorcyclists. The first, vulnerable_users, represents the aggregate count of such participants per collision. The second, has_vulnerable_user, is a binary indicator denoting whether at least one such user was present. This distinction is supported by prior knowledge demonstrating that vulnerable road user involvement is among the strongest predictors of fatal collision outcomes.

2) Darkness Exposure

A binary feature is_dark was constructed by detecting the substring "dark" in the Light field, capturing reduced-visibility conditions associated with higher collision severity.

3) Road Surface Hazard

The hazardous_surface binary feature flags records where Road_1_Surface_Condition is one of: loose snow, packed snow, wet, slush, or ice, conditions known to impair vehicle handling and increase stopping distance.

4) Environmental Hazard

Similarly, poor_environment flags adverse atmospheric conditions, including snow, rain, freezing rain, fog, and drifting snow, drawn from the Environment_Condition_1 field.

5) Traffic Control Absence

The no_control column is a binary indicator for collisions occurring at locations with no traffic control mechanism, a condition associated with higher unregulated vehicle speeds and conflict risk.

6) Multi-Vehicle Involvement

The multi_vehicle flags collisions involving more than one vehicle, capturing the energy transfer dynamics of multi-party impacts.

D. Interaction Feature

One interaction feature was constructed to capture compounded risk scenarios. vulnerable_x_dark is the product of has_vulnerable_user and is_dark, encoding collisions in which a vulnerable road user is present under dark conditions, a scenario where both reduced driver reaction time and

reduced pedestrian/cyclist visibility compound fatality risk simultaneously.

E. Categorical Encoding

Five categorical features — Initial_Impact_Type, Road_1_Surface_Condition, Environment_Condition_1, Light, and Traffic_Control, were encoded using one-hot encoding - `pd.get_dummies()` function with the first category dropped to avoid multicollinearity. This expanded the feature space from 22 pre-encoding columns to 60 total features. All resulting dummy variables were cast to integer type. The complete engineered dataset was exported for model training.

VI. MACHINE LEARNING APPROACH

Following Section V - Feature Engineering, the provided dataset assigned all attributes to numerical and applied the necessary encoding for machine learning. The objective was to apply various machine learning algorithms and determine the best algorithm for predicting Fatal Vs. Non-Fatal accidents in Ottawa.

Initially, the machine learning algorithms used were Logistic Regression, Random Forest, and XGBoost. These were chosen as they are well known algorithms when involving binary classification. The requirements for the project were to apply five models, thereby through research another boosting model, Gradient Boosting, was added. This was an optimal choice as it is known for handling imbalanced datasets. The Gradient Boosting was also chosen due to its ability to reduce errors in each iteration. The other model to be included was Linear Supervised Machine Learning (SVM), as SVM models are commonly used in classification problems [8]-[9].

Although SVM based algorithms are not optimal for imbalanced datasets, due to it maximizing the margin between points, it was chosen as a comparison model to determine if simpler or more complex models were better for handling the imbalance. Linear SVM was chosen over Non-Linear SVM as Linear SVM is faster to train and test and works well with high-dimensional data [9].

A. Imports

In order to apply the machine learning algorithms, the necessary imports were required. Basic Python libraries such as Pandas, Matplotlib, Numpy, and Seaborn were selected. Metric based libraries were imported from sklearn based on the requirements from Section II - Data Source. Each library associated with the five chosen models, Logistic Regression, Random Forest, XGBoost, Gradient Boosting, and Linear SVM were imported.

B. Data Imbalancing

As per Section II - Data Source, approximately 0.18% of the data is classified as 'Fatal'. This is an extremely imbalanced dataset. Thereby, to balance the dataset three balancing techniques were chosen; SMOTE, SMOTE-ENN, and ADASYN. All three listed balancing techniques are commonly applied to machine learning algorithms for balancing. Each technique was chosen for a separate reason. Synthetic Minority Over-Sampling Technique (SMOTE) applies oversampling by creating synthetic data points in turn to prevent overfitting. SMOTE-Edited Nearest Neighbors (ENN) is a variant of SMOTE for oversampling and ENN for cleaning. SMOTE-ENN removes ambiguous or noisy data

points from both classifications. Adaptive Synthetic Sampling (ADASYN) is another variant of SMOTE that focuses more on the minority samples that can be difficult to learn. [10]

C. Model Preparation

Due to the massive data imbalance, a dictionary and lists were applied in preparation for model evaluation. The objective was to evaluate different hyper parameters on each balancing technique against the test splits and basic parameters of the models. This in turn would give the best balancing technique and associated model for future hyper parameter tuning. This was done after trial and error to eliminate redundant repetition in the code, and to save time when evaluating.

The balancing techniques were defined in a list, each item contained a string name for labeling and the balancing technique. Each balancing technique was assigned a `random_state` of 42, `sampling_strategy` altered between 0.1 and 0.05, and `k_neighbors` was assigned to 3 when necessary. The models were defined in a dictionary with the models name and initializations, all models used the default parameters, with a few exceptions. These exceptions include `random_state` of 42 for all models, `max_iter` of 100 for Logistic Regression and Linear SVM, and `eval_metric` set to 'aucpr' for XGBoost. The different test_size were defined in a list, the splits used were 70/30, 80/20, and 85/15.

Lastly, the predicted column from the DataFrame, 'Classification_Of_Accident', was dropped for the X value and assigned only to the Y value.

D. Model Implementation

An empty results list was defined to store the evaluation metrics from each experiment, where each entry later becomes a dictionary of metrics. An empty predictions dictionary was used to store model predictions. The key is the experimented metrics, this consisted of the test split, balancing technique, and model. The value consisted of the predicted labels.

To evaluate each model, test split, and balancing technique, multiple for loops were added in. The split list was looped through first, from there the `train_test_split` function was used. The `random_state` was set to 42 for uniformity and stratify was set to y to have the same class distribution in both train and test sets.

The balancers were then looped through next. Resampling was applied to fit the balancer on the training data and generate the synthetic minority samples.

The models were then looped through and fit to learn from the balanced training dataset. A `test_pred` variable uses the `predict` function to predict the class label for the test dataset. Some models give different outputs, either probabilities or decision scores; these values are needed for calculating the Precision-Recall Area Under the Curve.

All necessary metrics are calculated, precision, recall, f1, matthews correlation coefficient, accuracy score, and true negative rate.

Each result was appended to the results list and the stored prediction to the predictions dictionary. The results were then cast to a DataFrame for further inspection.

VII. RESULTS AND DISCUSSION

The resulting DataFrame from Section VI d - Model Implementation was used to inspect which test split, balancing technique, and model combination had the best results. The best resulting combination is then used for further hyper parameter tuning to determine if there is a better model.

A. Model Evaluation

To evaluate the best balancing technique, test split, and model, the Matthew Correlation Coefficient (MCC) metric was used, as defined in Section II - Data Source. The resulting DataFrame from Section VI d - Model Implementation, was sorted based on the MCC and filtered for the top five results. MCC was chosen as the value to sort as it takes all four elements of the confusion matrix into account, compared to the Precision-Recall Area Under the Curve (PR-AUC) that evaluates how the model ranks positive samples. As seen in Fig. 15, the Gradient Boosting model performed the best, will SMOTE and SMOTE-ENN for good balancers. The parameters for the balancing techniques saw a dominance in `sampling_strategy = 0.1` and `k_neighbors = 3`.

Split	Balancer	Model	Precision	Recall	F1	PR_AUC	MCC	Accuracy	TNR	
198	0.15	smote_0.1_k3	Gradient Boosting	0.070175	0.153846	0.096386	0.028249	0.101416	0.994556	0.996146
18	0.30	smote_0.1_k3	Gradient Boosting	0.064000	0.156863	0.090909	0.020935	0.097610	0.994193	0.995746
108	0.20	smote_0.1_k3	Gradient Boosting	0.059701	0.117647	0.079208	0.018939	0.081433	0.994937	0.996564
172	0.20	smoteenn_0.05	XGBoost	0.076923	0.088235	0.082192	0.016624	0.080563	0.996353	0.998037
78	0.30	smoteenn_0.1_k3	Gradient Boosting	0.066667	0.098039	0.079365	0.021780	0.078781	0.995790	0.997455

Fig. 15. Top 5 Best Model Combinations.

As seen in Fig. 15, the top performing evaluation had a test split of 85/15, used a SMOTE balancing technique with `sampling_strategy = 0.1`, and `k_neighbors = 3`, and used the Gradient Boosting model. The top evaluation scored an MCC of approximately 0.1, a low value but expected for an extremely imbalanced dataset. Therefore, the top performing evaluation was chosen for hyper parameter tuning.

Each of the top five models, test split, and balancing technique were graphed to visualize the PR-AUC. However, the value 'PR_AUC' in the DataFrame (Fig. 15) takes the average. Therefore, to plot the entire graph the selected model name, balancing technique, and test split were extracted. The `train_test_split` function was applied again, the balancing technique was fit to resample, and the model was fit. The balancer name from the DataFrame was compared to the balancer dictionary, as discussed in Section V c - Model Preparation, to get the pre-defined balancing technique.

To plot the area under the curve, the sklearn metric, `PrecisionRecallDisplay` calls the `from_estimator` function to graph the PR-AUC (Fig. 16). The plot distribution is not greatly visible due to the imbalanced dataset.

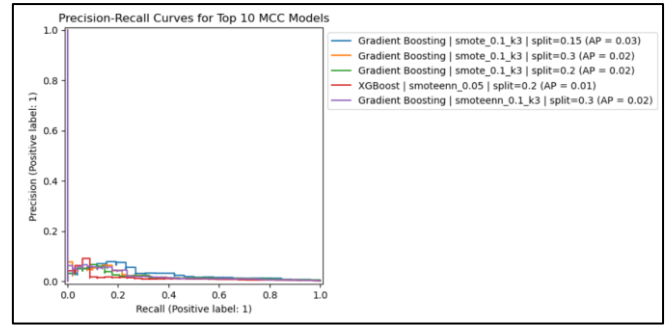


Fig. 16. Precision-Recall Area Under the Curve for Top 5 Models

B. Hyper Parameter Tuning

To apply Hyper Parameter tuning to the top performing evaluation, refer to Section VI a - Model Evaluation, the best balancer was defined. A `StratifiedKFold` was applied for cross validation as it preserves the percentage of samples for each class. The best balancer was `resample` to fit and the Gradient Boosting model was defined. A `param_grid` dictionary was used to loop through different iterations when applying a `GridSearchCV`. Within the grid search other parameters such as `n_jobs` were set to `-1` and `verbose` set to `2`. The former allows the program to use all available CPU cores and `verbose` displays the number of folds, candidates, and total fits.

The grid is then fitted based on the best balancer resampling, and the `best_params_` function is called to display the best finding. This is compared to the built in `best_estimator_` when estimating its findings.

The tuned model is then fit to train and evaluated on the test set. The test set displays all necessary metrics. The tuned Gradient Boosting model achieved a precision of 0.035 and recall of 0.059. The resulting F1-score of 0.044 and PR-AUC of 0.012 indicates a limited ability in identifying between Fatal and Non-Fatal accidents. Most notably the Mathews Correlation Coefficient (MCC) scored an approximate 0.043, this suggests a weak relationship between the predicted and actual classifications. However, the True Negative Rate (TNR) scored approximately 0.997, this suggests a high capability in identifying Non-Fatal accidents.

In comparison to the untuned model, Fig. 15, the MCC was approximately 0.10 and PR-AUC was 0.028. Therefore, the tuned model performed worse. The difference in performance may be an indicator of the extreme class imbalance and limited number of Fatal samples in the dataset.

VIII. IMPACTS AND APPLICATIONS

One of the most significant impacts of this project lies in its application to emergency response optimization. Emergency services such as police, fire, and ambulance units must make rapid decisions when responding to 911 calls. By leveraging predictive insights from this model, emergency responders could better assess the potential severity of a collision in real time. This would enable more informed decisions regarding the type and number of resources to deploy, ultimately improving response efficiency and potentially saving lives by ensuring that high-risk incidents receive immediate and adequate attention.

Furthermore, the findings from this project can support data-driven decision-making in urban planning and public safety policy. By identifying patterns and high-risk conditions associated with fatal accidents, city officials can implement targeted interventions such as improving road infrastructure, enhancing lighting in high-risk areas, and increasing enforcement in dangerous zones. Overall, this project demonstrates how machine learning can be applied to real-world safety challenges, contributing to more effective emergency response strategies and safer road environments in Ottawa.

IX. CONCLUSION

In conclusion, the study investigated the effectiveness of multiple machine learning algorithms on an extremely imbalanced dataset aimed to predict Fatal Vs. Non-Fatal car accidents in Ottawa. Five models were evaluated, such as Logistic Regression, Random Forest, XGBoost, Gradient Boosting, and Linear SVM. In an attempt to balance the dataset for modeling, resampling techniques, such as SMOTE, SMOTE-ENN, and ADASYN, were tested across each model and train-test splits.

The experiment resulted in the Gradient Boosting model combined with SMOTE with a sampling strategy of 0.1 and $k_neighbors$ of 3 on an 85/15 train-test split performed the best. The performance was based on the Matthews Correlation Coefficient (MCC), with the highest value of 0.10 approximately. The low results across varying metrics, aside from accuracy, were expected due to the extreme imbalance.

Moreover, hyper parameter tuning was applied to the top result. It was configured using cross-validation and GridSearchCV. As a result, the tuned model did not have any significant improvements in its prediction. It continued to struggle with accurately identifying the Fatal accident class.

Therefore, this project highlights the challenges associated with predicting extremely imbalanced datasets. Even with the inclusion of resampling techniques and hyper parameter tuning, accurately detecting a rare fatal accident with a limited number of samples remains difficult.

X. FUTURE WORK

Future work should focus on the primary limitation of this project, which is the extreme class imbalance in fatal collision records. There were only 170 fatal samples out of 94,406 total observations. Datasets from other municipalities or provincial sources should be investigated to increase the representation of fatal events and improve the model's ability to learn from the minority class.

Additionally, neural networks may be explored and implemented to capture more complex, non-linear relationships within the data.

Lastly, while this study identified high-frequency collision locations through EDA, further research could apply advanced spatial techniques such as geospatial clustering to improve the identification of high-risk areas.

REFERENCES

- [1] J. Pringle, "6% jump in collisions on Ottawa roads so far this year," CTVNews, Feb. 26, 2026. <https://www.ctvnews.ca/ottawa/article/6-jump-in-collisions-on-ottawa-roads-so-far-this-year/> (accessed Mar. 23, 2026).
- [2] City of Ottawa, 2026, "Traffic Collision Data," Open Data Ottawa. [Online]. Available: <https://open.ottawa.ca/datasets/ottawa::traffic-collision-data/about> (accessed Apr. 13, 2026).
- [3] J.R. Tiwari, "Advanced Evaluation Metrics for Imbalanced Classification Models," CueNex, Feb. 09, 2023. <https://medium.com/cuenex/advanced-evaluation-metrics-for-imbalanced-classification-models-ee6f248c90ca> (accessed Apr.13, 2026)
- [4]]Leonidas Gorgo, "Understanding ROC-AUC and PR-AUC," Medium, Aug. 27, 2025. <https://leonidasgorgo.medium.com/understanding-roc-auc-and-pr-auc-c4881af45a16> (accessed Apr. 13, 2026)
- [5] Iván Palomares Carrascosa, "ROC AUC vs Precision-Recall for Imbalanced Data - MachineLearningMastery.com," MachineLearningMastery.com, Sep. 09, 2025. <https://machinelearningmastery.com/roc-auc-vs-precision-recall-for-imbalanced-data/> (accessed Apr.13 2026)
- [6] "Welcome To Zscaler Directory Authentication," Kaggle.com, 2026. <https://www.kaggle.com/discussions/general/253378> (accessed Apr. 13, 2026)
- [7] Anishnama, "Matthews Correlation Coefficient(MCC) one of the best metric when 2 classes are imbalanced," Medium, Jan. 16, 2023. <https://medium.com/@anishnama20/matthews-correlation-coefficient-mcc-one-of-the-best-metric-when-2-classes-are-imbalanced-c0318ac68c21> (accessed Apr.13 2026)
- [8]]Shivam Chhetry, "Handling Imbalanced Dataset In Machine Learning.," Medium, May. 03, 2021. <https://shivamkc01.medium.com/handling-imbalanced-dataset-in-machine-learning-9ac075787e07> (accessed Apr. 14, 2026).
- [9] [1]IBM, "Support Vector Machine," IBM, Dec. 12, 2023. <https://www.ibm.com/think/topics/support-vector-machine> (accessed Apr. 14 2026)
- [10] T. Tang, "Class Imbalance Strategies — A Visual Guide with Code," Medium, Apr. 24, 2023. <https://medium.com/data-science/class-imbalance-strategies-a-visual-guide-with-code-8bc8fae71e1a> (accessed Apr. 14, 2026)